2. Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 1 in Column 2
∴ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
∴ The condition that for each column having a leading entry, all posns. below the leading entry must contain 0 is True.

4. All leading entries are 1. This condition is True.

5. Row 1 Column 2 = 0
∴ The condition that for each column having a leading entry, all posns. above the leading entry must contain 0 is True.

∴ The matrix $\begin{bmatrix} 1 & 0 & -5 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in reduced row echelon form. □

By defn, there is no pivot in Column 3. ∴ By defn., x_3 is a free variable.
The corresponding equations are: ① $x_1 - 5x_3 = 0$ ② $x_2 + 2x_3 = 0$
∴ $x_1 = 5x_3$, $x_2 = -2x_3$, x_3 is a free variable.
In parametric vector form, the solution is: $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$

In Exercises 7–12, describe all solutions of $Ax = 0$ in parametric vector form, where A is row equivalent to the given matrix.

7. $\begin{bmatrix} 1 & 3 & -3 & 7 \\ 0 & 1 & -4 & 5 \end{bmatrix} \xrightarrow[3 \times \text{Row } 2]{\text{Row }1 = \text{Row }1 -} \begin{bmatrix} 1 & 0 & 9 & -8 \\ 0 & 1 & -4 & 5 \end{bmatrix}$

Verification of whether the final matrix is in rref form

1. Row 1 and Row 2 are non-zero rows. ∴ The condition that all non-zero rows must be above the zero rows is vacuously True.